

DEPAUL UNIVERSITY

Week 3 Deliverable – Interactive Dashboard & Insight Report

Interactive Dashboard File & Demo Video

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Interactive Dashboard File (ZIP File):

<https://drive.google.com/file/d/1O7iW-Ect2LGQY0zbZJYSutuWMVUYRPF4/view>

Tool: Microsoft Power BI Desktop

Dashboard Demo Video:

<https://go.screenpal.com/watch/cO1rIQnuKtY>

1. Executive Summary

This report documents a four-page interactive Power BI dashboard built on DePaul University's cleaned applicant dataset (7,543 records, 5,000 unique applicants, May 2023–June 2025). The dashboard spans four views: Executive Summary, Program Analysis, Geographic Analysis, and Recruitment Analysis, each addressing a distinct stakeholder question and connected through shared slicers and cross-filtering.

Metric	Value	Metric	Value
Total Applicants	5,000	Admitted	5,789 records
Slate Form Filled	4,336 (99.7%)	Study Group Share	13.5%
Countries Represented	20+	Top Program	Business Analytics MS

2. Dashboard Design Documentation

2.1 Layout & Page Structure

Pg	Name	Key Visuals
1	Executive Summary	KPI cards, trend line, source doughnut, college bar chart, date slicer
2	Program Analysis	Program ranking bar, term-over-term columns, college×term matrix
3	Geographic Analysis	Filled world map, country table, top-10 bar, cross-filter
4	Recruitment Analysis	Funnel chart, outcome bars, caller table, campaign slicer

2.2 Why Each Visual Was Chosen

Visual	Rationale
KPI Cards	Instant numeric read for leadership without chart interpretation.
Line Chart	Best format for a continuous time series; shows seasonal trend direction clearly.
Doughnut Chart	Communicates Study Group vs. Non-Study Group part-to-whole proportion at a glance.
Filled Map	Encodes applicant volume spatially through color, more intuitive than a country list.
Funnel Chart	Most intuitive way to show a sequential conversion process and where it narrows.
Matrix + Conditional Formatting	Cross-references two categories (college × term); color gradient flags hot spots instantly.

2.3 Interactivity & Data Flow

Filters, slicers, and cross-filtering

- Date slicer (Admit_Date) on Page 1 filters all KPI cards and charts together by quarter or year.
- Program Name and Term slicers (from the split Intake field) let any page isolate a single program or cohort.
- Default Power BI cross-filtering links every visual on a page — e.g., clicking a country on the map filters the program bar chart to that country's applicants.
- Seven DAX measures (Total Applicants, Total Admitted, Admission Rate, Slate Filled, Contacted, Follow Up Count, Missing Transcript) recalculate live as filters change, keeping every page synchronized from one underlying table.

2.4 Data Preparation Summary

Before building, the raw export was cleaned in Power Query: two empty trailing columns were removed; Admit_Date, Start_Date, and contact-date fields were converted to proper Date types; the Intake field was split into Program_Name and Term using a year-season pattern (since it mixed MS, MBA, MA, and PhD delimiters); and Country/College text values were standardized to remove duplicate spellings.

3. Insight Report

Key trends and patterns identified across the four dashboard pages, each tied to a specific visualization.

3.1 Program & Geographic Findings

Finding	Detail
Business Analytics MS leads demand	1,080 applicants, ahead of Computer Science MS (1,055); demand is steady across Fall and Winter, not just seasonal.
AI program is accelerating	Artificial Intelligence applications roughly doubled term-over-term from 2023 to 2024 — a growth signal worth capacity planning for.
Jarvis College dominates	Computing and Digital Media draws 50.8% of first-choice applications vs. 33.5% for Kellstadt Business — STEM clearly drives the pipeline.
India is 61% of the pipeline	4,617 of 7,543 applicants originate from India, creating both a deep talent source and a geographic concentration risk.
Africa is an emerging market	Ghana (440) and Nigeria (430) combined exceed the entire U.S. domestic applicant pool (764) — an underleveraged growth region.

3.2 Recruitment Funnel Findings

Finding	Detail
Transcript collection is the top blocker	1,177 applicants are stalled on 'Missing Transcript' — the largest single drop-off cause in the funnel.
Follow-up is the biggest recoverable segment	1,375 applicants sit in 'Follow up' status — engaged but not yet converted, the single largest yield opportunity.
Three counselors carry 99% of outreach	Jagdeep (2,085), Swarna (1,214), and Ayesha (855) handle nearly all calls — a concentration risk if any is unavailable.
Slate form completion is strong	99.7% completion suggests counselor follow-through is effective once an applicant engages with the process.

4. Actionable Recommendations

4.1 Recruitment Targeting

- Diversify beyond India by launching targeted campaigns in Pakistan, Kenya, and other markets showing demonstrated but smaller-scale interest.
- Invest in Ghana and Nigeria with dedicated outreach, since these two countries already outproduce the entire U.S. domestic pipeline.
- Offer stronger Winter/Spring incentives to balance the 35% Fall-term concentration and smooth staff workload across the year.

4.2 Counselor Engagement

- Cross-train additional staff to reduce reliance on three counselors who handle nearly all outreach calls.
- Build a structured 30/60/90-day follow-up sequence to convert the 1,375 applicants currently stalled at 'Follow up'.
- Create a proactive transcript-reminder workflow triggered at application submission to reduce the 1,177 'Missing Transcript' cases.

4.3 Admission Outcomes

- Expand AI program capacity and marketing ahead of demand, given its term-over-term doubling in applications.
- Track Business Analytics selectivity to ensure admission standards keep pace with its leading applicant volume.
- Evaluate Study Group ROI with a cost-per-enrolled-student analysis, since it sources 13.5% of applicants through a paid partnership channel.

5. Dashboard Screenshots

Inserted annotated screenshots of each dashboard page below. In Power BI Desktop: File > Export > Export to PDF, or capture each page individually.

Fig.	Dashboard Page	Key Components Shown
1	Executive Summary	KPI cards, trend line, source donut, college bar chart
2	Program Analysis	Program ranking, term comparison, collegexterm matrix
3	Geographic Analysis	Filled world map, country table, top-10 bar chart
4	Recruitment Analysis	Funnel chart, outcome bars, caller table, slicers

Figure 1 — Executive Summary

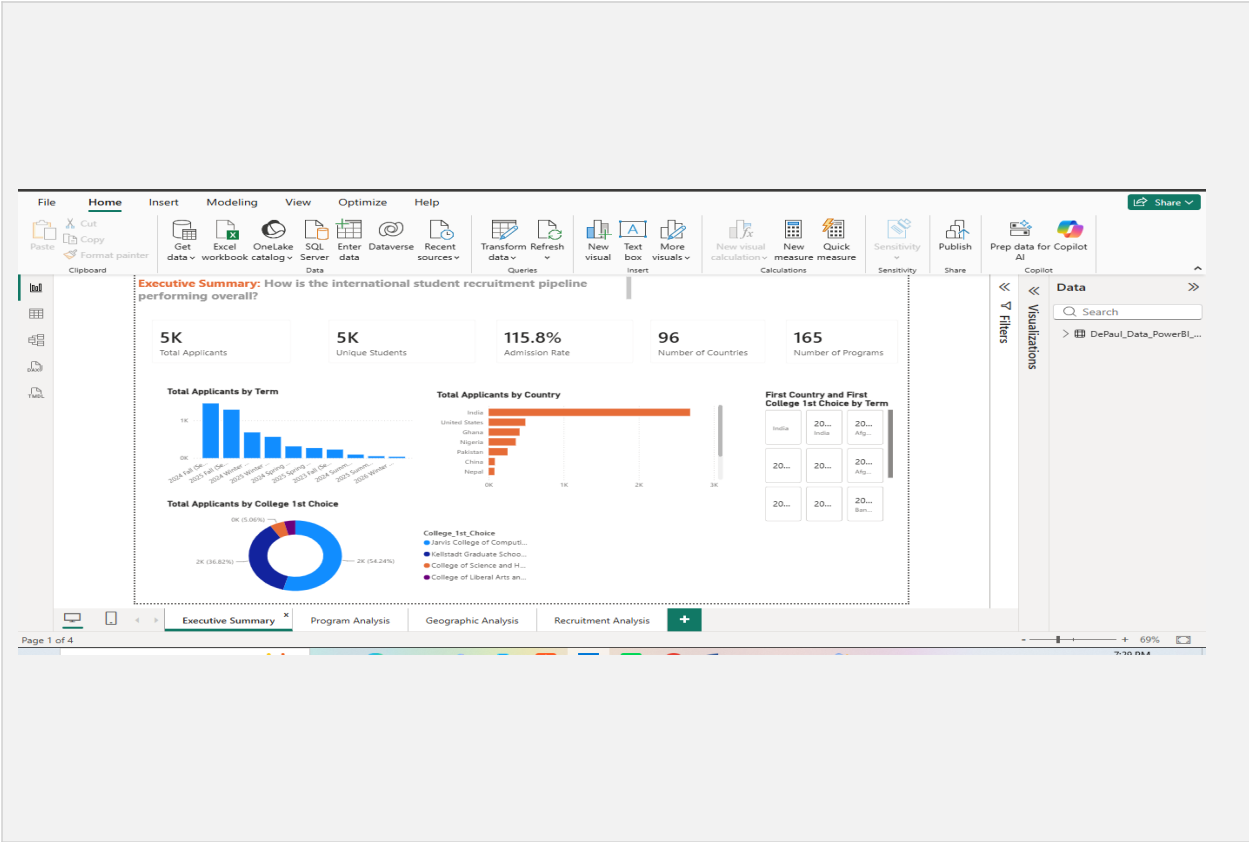


Figure 2 — Program Analysis

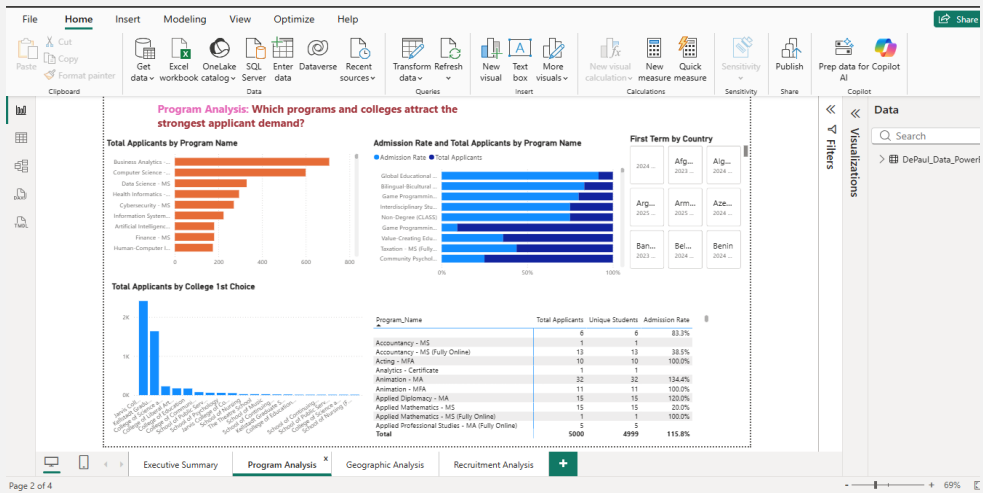


Figure 3 — Geographic Analysis

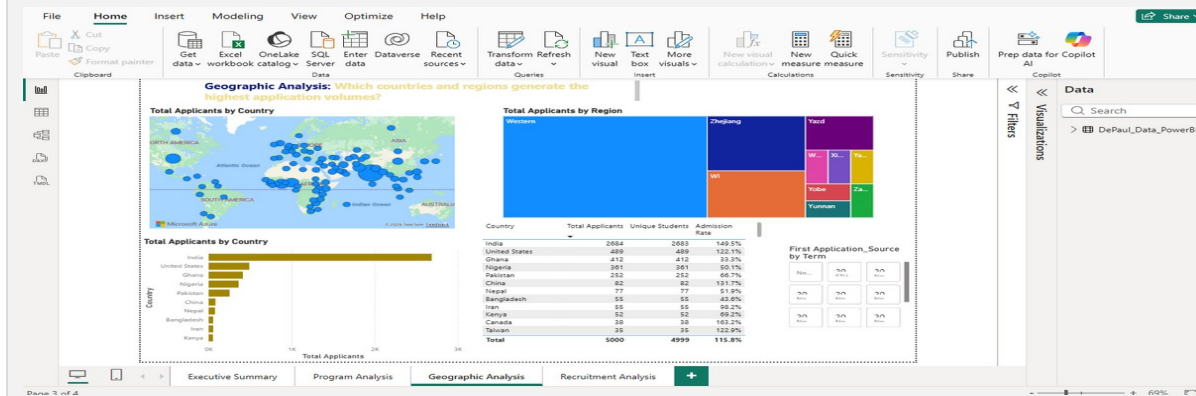
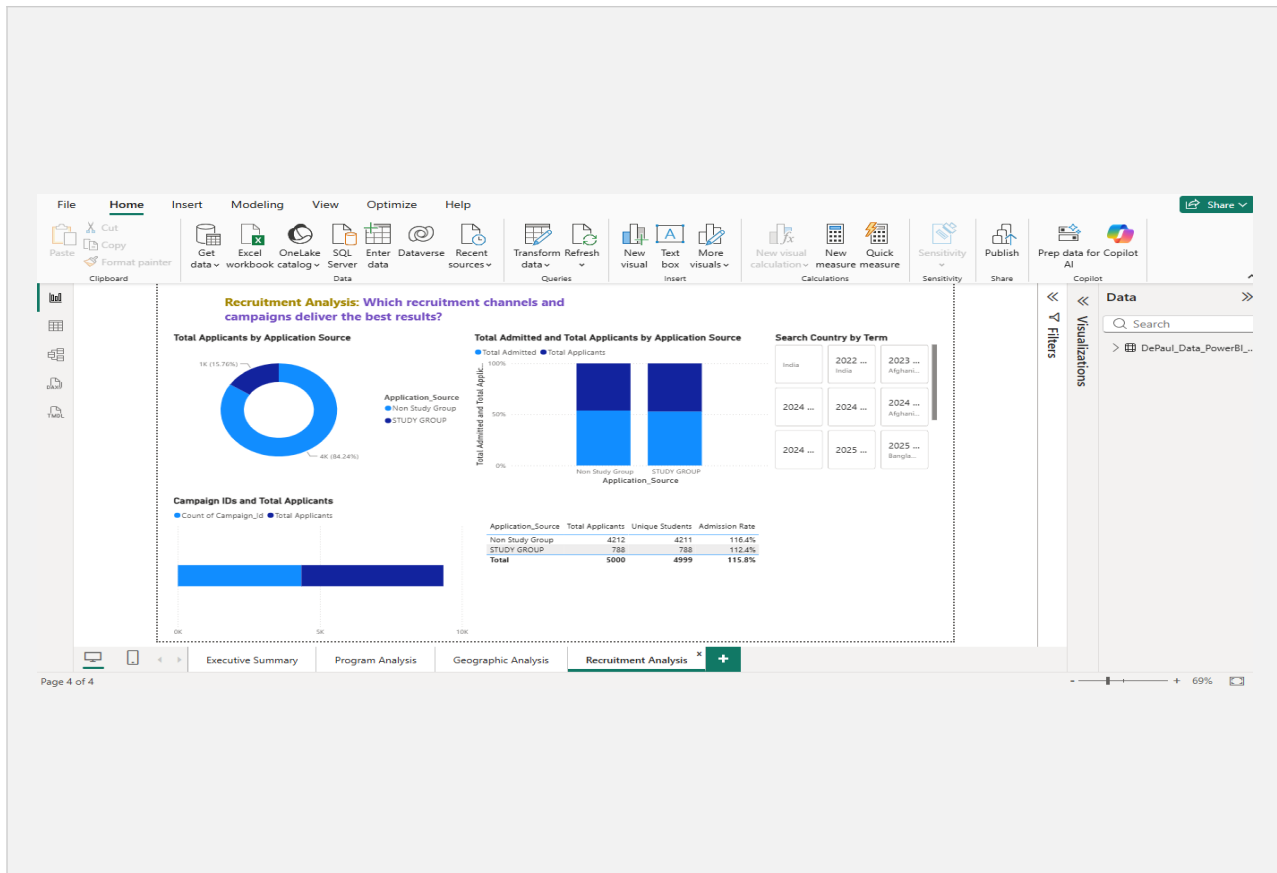


Figure 4 — Recruitment Analysis



To view our fully functional dashboard, click this link to our interactive dashboard zip file [here](#).

6. References & Tools

- Microsoft Power BI Desktop (2024). Data visualization and business intelligence platform. Microsoft Corporation.
- Power Query M Language and DAX (Data Analysis Expressions), used for data cleaning and calculated measures.
- DePaul University Applicant Dataset (Cleaned), internal dataset provided for the Week 3 deliverable.